



MAE101-PT1 - MAE101

Mathematics Engineering (Trường Đại học FPT)



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Câu 1. Given $f(u) = u^2$, $g(x) = x^5 + 2$, find $(f \circ g)'(1)$

A. 6

B. 15

C. 30

D. None of the other choices is correct

E. -30

Câu 2. Find the relative extrema, if they exist, of the function $f(x) = \frac{8x}{x^2+1}$

A. Relative minimum at $(-1, -4)$, relative maximum at $(1, 4)$

B. Relative minimum at $(-1, -4)$, relative maximum at $(0, 0)$

C. Relative maximum at $(-1, -4)$, relative minimum at $(1, 4)$

D. Relative maximum at $(0, 0)$, no relative minimum

E. None of the other choices is correct

Câu 3. Evaluate the limit, if it exists: $\lim_{x \rightarrow 1} \frac{\sqrt{x^2+3x}-2}{x-1}$

A. 0

B. 5/2

C. 5/4

D. -5/2

E. None of the other choices is correct

F. -5/4

Câu 4. A table of values for f , g , f' , g' is given:

x	f(x)	g(x)	f'(x)	g'(x)
0	1	1	2	-2
1	0	2	3	-1
2	4	-1	5	6

Find $h'(1)$ if $h(x) = g(f(x))$.

A. 0

B. -6

C. None determined

D. -2

E. None of the other choices is correct

F. -5

Câu 5. Find dy/dx by implicit differentiation $xy^2 = 4$.

A. None of the other choices is correct

B. $-2y/x$

C. $2x/y$

D. $-y/(2x)$

E. $x/(2y)$

Câu 6. Use the Simpson's Rule with $n = 4$ to approximate the integral:

$$\int_1^5 x \cos x dx$$

A. None of the other choices is correct

B. -5.92

C. -6.26

D. -3.43

E. -6.89

Câu 7. Simplify the quotient $\frac{f(x+h)-f(x)}{h}$ for $f(x) = \frac{1}{x}$

(i) $\frac{-1}{(x-h)x}$ (ii) $\frac{1}{(x+h)x}$ (iii) $\frac{-1}{(x+h)x}$ (iv) $\frac{-1}{(x+h)(x-h)}$

A. (iii)

B. (iv)

C. (i)

D. (ii)

E. None of the choices is correct

Câu 8. Find the vertical asymptotes to the graph of the following function:

$$f(x) = \frac{x-4}{6-\sqrt{x}-x}$$

A. $x = -4$ and $x = 6$

B. $x = 4$ and $x = 6$

C. $x = 4$ and $x = -6$

D. Does not exist

Câu 9. Find the average value of the function $f(x) = 7x + 5$ on the interval $[1;4]$

A. 7

B. None of the other choices is correct

C. 22.5

D. 2

E. 14

F. 4

Câu 10. Express the limit as a definite integral on the given interval.

$$\lim_{n \rightarrow \infty} \sum_{i=1}^n \cos^2(2\pi x_i^*) \Delta x$$

(i) $\int_0^1 \cos^2(2\pi) dx$ (ii) $\int_0^1 \cos^2\left(\frac{2\pi}{x}\right) dx$ (iii) $\int_{-1}^1 \cos^2(2\pi x) dx$ (iv) $\int_0^1 \cos^2(2\pi x) dx$

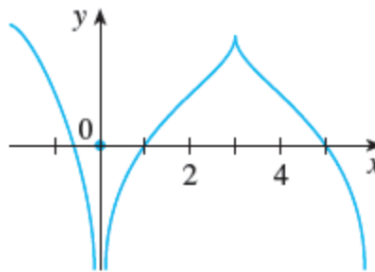
- A. (iv)
- B. (iii)
- C. (i)
- D. None of the other choices is correct
- E. (ii)

Câu 11. A particle is moving with the given data. Find the position function $s(t)$ of the particle.

$$v(t) = 3\sin t - 5\cos t, \quad s(\pi) = 1$$

- A. $s(t) = 3\cos t + 5\sin t - 2$
- B. $s(t) = 3\cos t - 5\sin t - 2$
- C. $s(t) = -3\cos t - 5\sin t + 4$
- D. None of others
- E. $s(t) = 3\cos t - 5\sin t - 2$

Câu 12. The graph of $f(x)$ is given. State the numbers at which $f(x)$ is not differentiable



- A. 0; 1; 3; 4
- B. 0; 3
- C. 3
- D. 0; 2; 4
- E. None of the other choices is correct

Câu 13. Using the Midpoint Rule with $n = 4$ to approximate

$$\int_1^5 (x + 0.5) dx$$

- A. 14
- B. 12
- C. 16
- D. None of the choices is correct
- E. 20

Câu 14. Which of the following integrals is divergent?

$$(i) \int_1^{\infty} \frac{2010}{5x^2} dx \quad (ii) \int_1^{\infty} \frac{1}{\sqrt{x^3}} dx \quad (iii) \int_1^{\infty} \frac{1+xe^{-2x}}{x} dx$$

- A. (iii)
- B. (i)

C. (ii)

D. None of the other choices is correct

Câu 15. Find $f(x)$ such that $f'(x) = x^2 - 11x + 7$ and $f(0) = 3$

(i) $f(x) = \left(\frac{1}{3}\right)x^3 - \left(\frac{11}{2}\right)x^2 + 6x + 1$

(iii) $f(x) = \left(\frac{1}{3}\right)x^3 - \left(\frac{11}{2}\right)x^2 + 7x + 3$

(ii) $f(x) = \left(\frac{1}{3}\right)x^3 - 12x^2 + 7x + 3$

(iv) $f(x) = \left(\frac{1}{3}\right)x^3 - 12x^2 + 7x + 1$

A. None of the other choices is correct

B. (ii)

C. (iv)

D. (iii)

E. (i)

Câu 16. Find $y'(0)$ for $y = (3x^2 + 5x + 1)^{3/2}$

A. $5/2$

B. $15/2$

C. None of the choices is correct

D. $1/2$

E. $3/2$

Câu 17. Simplify the quotient $\frac{f(x+h)-f(x)}{h}$ for $f(x) = \frac{x^2}{2}$

A. $(2x + h)/2$

B. $(x - 2h)/2$

C. None of the other choices is correct

D. $(-2x - h)/2$

E. $x/2$

Câu 18. Find all the numbers that satisfy the conclusion of Rolle's Theorem

$$f(x) = x^2 - 5x + 1; \quad [0; 5]$$

A. 5 and 2

B. None of the other choices is correct

C. 2 and 5

D. $5/2$

E. 0 and 5

F. $-5/2$

Câu 19. At which points on the curve $y = 5x + 10x^3 - 3x^5$ do the tangent lines have largest slope?

A. (0, 0) and (1, 12)

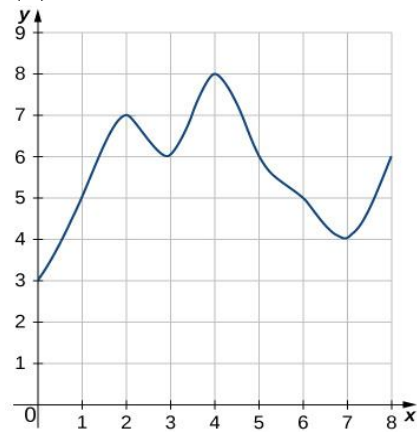
B. Does not exist

C. (0, 0)

D. $(-1, -12)$ and $(1, 12)$

E. $(-1, -12)$ and $(0, 0)$

Câu 20. Given the curve of $f(x)$ on the interval $[0,8]$, compute the right Riemann sum R_8



A. 47

B. 45.5

C. 44

D. None of the other choices is correct